

Xerosols

Description:

Xerosols are soils of arid and semi-arid regions characterized by very limited leaching and weak horizon development due to low and erratic rainfall. They typically form on alluvial, colluvial, or aeolian parent materials in dry climates with sparse vegetation cover.

The name “Xerosol” derives from the Greek *xeros*, meaning “dry,” reflecting their dryland formation conditions. These soils are usually light-colored (pale brown to greyish brown), coarse-textured (sandy to loamy), and low in organic matter. Calcium carbonate and gypsum often accumulate in the subsoil, forming weak calcic or gypsic horizons.

Drainage:

- **Wet season:** Well-drained to excessively drained due to coarse texture and low rainfall; minimal surface runoff.
- **Dry season:** Very dry and hard; low moisture-holding capacity leads to drought stress.

Landscape:

Commonly found on gently sloping to undulating plains, alluvial fans, and piedmonts of arid and semi-arid zones. In Kenya, Xerosols occur in areas receiving less than 500 mm of annual rainfall — notably in parts of Turkana, Marsabit, Garissa, and northern Baringo counties.

Depth:

Shallow to moderately deep (30–100 cm), depending on parent material and erosion. Rooting depth may be limited by compacted calcareous or stony layers.

Workability:

- **When moist:** Easy to cultivate and friable, allowing for good seedbed preparation.
- **When dry:** Becomes hard and crusted; surface sealing and wind erosion are common. Proper timing of tillage is essential to avoid crusting and loss of structure.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low to moderate (5–15 cmol(+)/kg) due to sandy texture and low organic matter.
- **Base saturation:** High (>70%) with calcium and magnesium dominating the exchange complex.
- **Nutrients:** Nitrogen and phosphorus are typically deficient; potassium generally adequate. Micronutrient deficiencies (Zn, Fe) can occur due to alkalinity.
- **Organic matter:** Very low (<1.5%), limiting biological activity and aggregation.

pH:

Neutral to moderately alkaline (7.0–8.5), influenced by accumulation of calcium carbonate and low leaching.

Management:

- Incorporate organic matter (compost, manure, green manure) to improve soil structure, water-holding capacity, and nutrient retention.
- Use mulching and conservation tillage to reduce evaporation and erosion.
- Apply nitrogen and phosphorus fertilizers in split doses for better efficiency.
- Establish windbreaks or shelterbelts to minimize wind erosion.
- Use drip or furrow irrigation and water-harvesting structures to supplement moisture.
- Avoid overgrazing and maintain vegetative cover to prevent surface sealing and desertification.

Suitable Crops:

Best suited for drought- and salt-tolerant crops such as sorghum, millet, cowpeas, pigeon peas, and green grams under rainfed conditions. Under managed irrigation, crops like maize, tomatoes, onions, and melons can be grown profitably.